Advanced Optics Laboratory (PH4201)

Experiment: 02

Determination of Polarization properties of Spatial Light Modulator (SLM) using Stokes Vector Polarimetry

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1 | Introduction

Optical polarimetry, ever since its discovery, has played important roles in gaining fundamental understanding of the nature of electromagnetic waves and answering some of the key questions related to the physics of light. Traditional polarimetry has also long being pursued for numerous practical applications in various branches of science and technology. Quantification of protein properties in solutions, testing of chiral (asymmetric) purity of pharmaceutical drugs, remote sensing in meteorology and astronomy, optical stress analysis of structures, and crystallography of biochemical complexes are just a smattering of its diversified uses. This particular polarimetry experiment involves determination of two important polarization characteristics of medium, namely, the linear retardance and optical rotation (defined subsequently), which has many practical applications.

2 | Objective

The main ojective of this experiment is to determine of linear retardance and Optical rotation of a Spatial Light Modulator (SLM) measuring various Stokes vector elements

3 | Equipment Required

The following equipments are required for this experiment

- Diode laser (650 nm)
- Spatial Light Modulator (SLM)
- Two Polarizer with mounts
- Photo-detector
- Optical table and mounts
- Computer with software for data acquisition and analysis
- Power supply for the SLM
- Glass plate with stage having rotational circular scale(in degree)

4 | Theory

4.1 | Basics of Polarization

Christian Huygens was the first to suggest that light was not a scalar quantity based on his work on the propagation of light through crystals; it appeared that light had 'sides' in the words of Newton. This vectorial nature of light is called polarization [1].

In a Cartesian system the components $u_x(\mathbf{r}, t)$ and $u_y(\mathbf{r}, t)$ are said to be the transverse components, and the component $u_z(\mathbf{r}, t)$ is said to be the longitudinal component when the propagation is in the z direction. Thus we can write

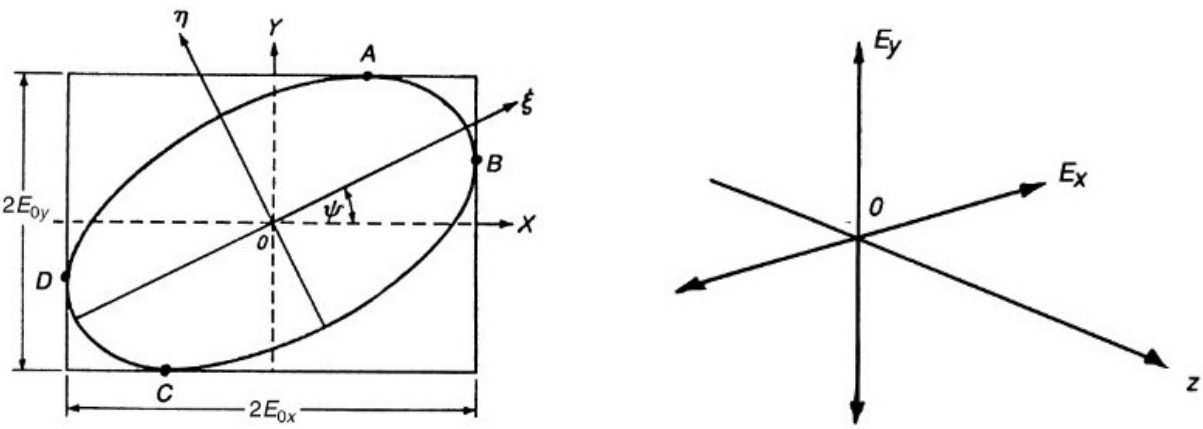
$$u_x(\mathbf{r}, t) = u_{0x} \cos(\omega t - \mathbf{k} \cdot \mathbf{r} + \delta_x), \tag{4.1a}$$
$$u_y(\mathbf{r}, t) = u_{0y} \cos(\omega t - \mathbf{k} \cdot \mathbf{r} + \delta_y), \tag{4.1b}$$
$$u_z(\mathbf{r}, t) = u_{0z} \cos(\omega t - \mathbf{k} \cdot \mathbf{r} + \delta_z). \tag{4.1c}$$

Experimentally found that light consisted only of the transverse components Equation (4.1a) and Equation (4.1b). If we take the direction of propagation to be in the z direction, then the optical field in free space must be described only by Equation (4.1a) and Equation (4.1b). where u_{0x} and u_{0y} are the maximum amplitudes and δ_x and δ_y are arbitrary phases. These two Equation (4.1a) and Equation (4.1b) are are spoken of as the polarized or polarization components of the optical field. Now, the transverse components are represented by

$$E_x(z,t) = E_{0x}\cos(\tau + \delta_x), \tag{4.2a}$$
$$E_y(z,t) = E_{0y}\cos(\tau + \delta_y), \tag{4.2b}$$

where $\tau = \omega t - kz$ is the propagator. The subscripts x and y refer to the components in the x and y directions. E_{0x} and E_{0y} are the maximum amplitudes, and δ_x and δ_y are the phases, respectively. As the field propagates, $E_x(z,t)$ and $E_y(z,t)$ give rise to a resultant vector. This vector describes a locus of points in space, and the curve generated by those points will now be derived. In order to do this Equation (4.2a) and Equation (4.2b) are written as

$$\frac{E_x}{E_{0x}} = \cos\tau\cos\delta_x - \sin\tau\sin\delta_x, \tag{4.3a}$$
$$\frac{E_y}{E_{0y}} = \cos\tau\cos\delta_y - \sin\tau\sin\delta_y. \tag{4.3b}$$



(a) An elliptically polarized wave and the polarization ellipse

(b) Propagation of the transverse optical field

Figure 4.1: Polarization of light

Hence,

$$\frac{E_x}{E_{0x}}\sin\delta_y - \frac{E_y}{E_{0y}}\sin\delta_x = \cos\tau\,\sin(\delta_y - \delta_x), \tag{4.4a}$$
$$\frac{E_x}{E_{0x}}\cos\delta_y - \frac{E_y}{E_{0y}}\cos\delta_x = \sin\tau\,\sin(\delta_y - \delta_x). \tag{4.4b}$$

Squaring (4.4a) and (4.4b) and adding gives

$$\frac{E_x^2}{E_{0x}^2} + \frac{E_y^2}{E_{0y}^2} - 2\frac{E_x}{E_{0x}}\frac{E_y}{E_{0y}}\cos\delta = \sin^2\delta, \tag{4.5a}$$
$$\delta = \delta_y - \delta_x. \tag{4.5b}$$

Equation (4.5a) is recognized as the equation of an ellipse and shows that at any instant of time the locus of points described by the optical field as it propagates is an ellipse (see Figure 4.1a). This behavior is spoken of as optical polarization, and Equation (4.5a) is called the *polarization ellipse* [1].

Also, we can write some more parameters of ellipse in terms of δ and α where $\tan\alpha = E_{0y}/E_{0x}$

$$\tan 2\psi = (\tan 2\alpha)\cos\delta, \qquad 0 \le \psi \le \pi, \tag{4.6a}$$
$$\sin 2\chi = (\sin 2\alpha)\sin\delta, \qquad -\frac{\pi}{4} \le \chi \le \frac{\pi}{4}. \tag{4.6b}$$

4.2 | Stokes Vector and Muller Matrix

5 | Set up and Procedure

This experiment has two parts. In the first part, we did measure the angles of polarizers (P1 and P2) for Horizontal polarization. And in the later part we did Stokes polarimetry for SLM.

Part 1

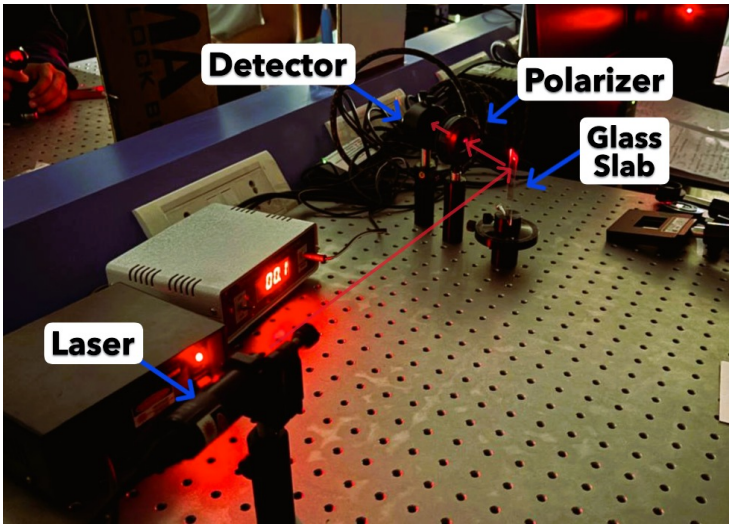


Figure 5.1: Setup to find the axis of the first polarizer (P1) using Brewster's angle approch

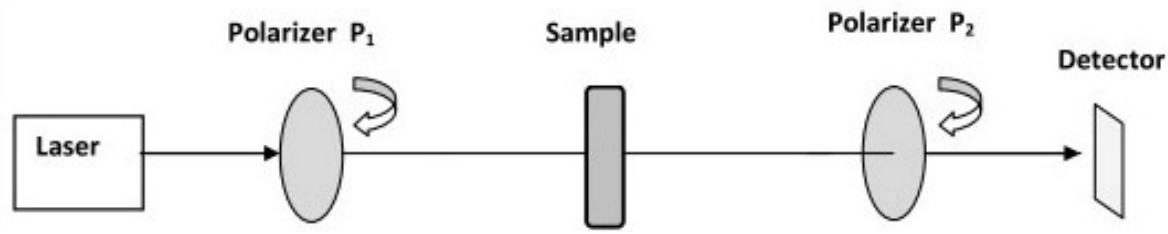


Figure 5.2: Schematic of the final setup

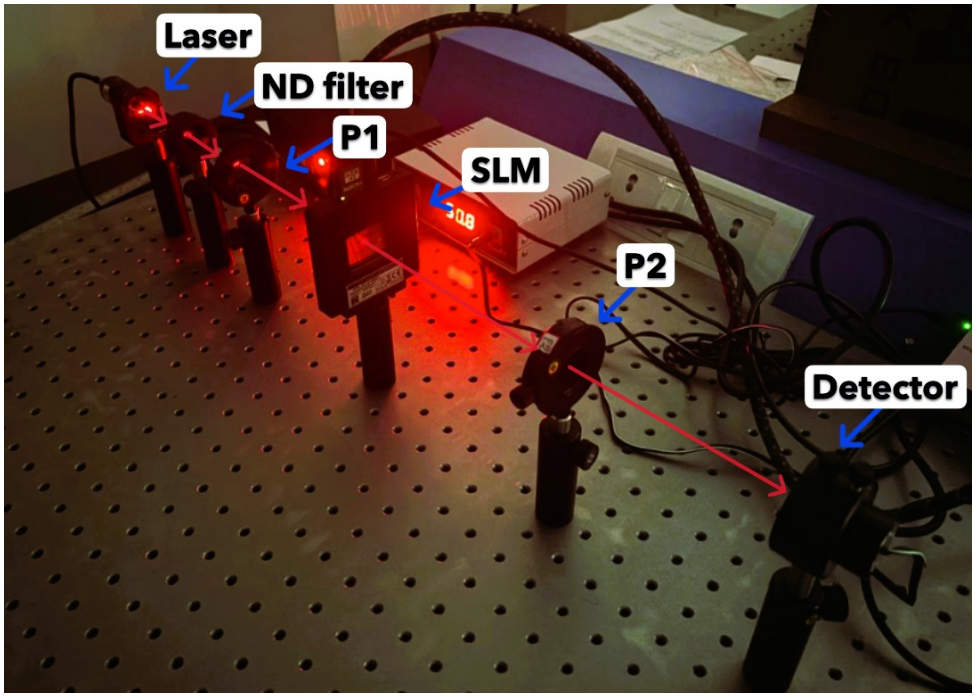


Figure 5.3: Final setup to find the Stokes measurement

6 | Data analysis and calculation

7 | Conclusion

8 | References

[1] Dennis H. Goldstein. *Polarized Light, Revised and Expanded.* CRC Press, Boca Raton, FL, 2003.